

Sangyun Lee

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EDUCATION

Keimyung University, Digipen institute of technology | Daegu, Korea & Redmond, WA, USA

B.S. in Computer Science | GPA: 3.8 / 4.5

Relevant Courses: Computer Graphics, Linear Algebra, Data Structures & Algorithms, Operating Systems

TECHNICAL SKILLS

Programming Languages: C++, GLSL

Graphics API & Tools: OpenGL 4.6

Interpersonal Skills: Solution-Oriented, Adaptable, Detail-Oriented

PROJECTS

Custom Game Engine | Engine Programmer

2025

- Developed a custom game engine from scratch using OpenGL 4.6 and GLSL shader programming, implementing a full rendering pipeline with advanced visual techniques
- Integrated rendering features including MSAA anti-aliasing and post-processing effects such as Fog, Toon Shading, Shadow Mapping, and gradient-based techniques to improve visual fidelity
- Designed hardware-friendly architecture with a low-level understanding of the C++ compiler and linker pipeline, achieving performance optimization through cache-efficient data layouts
- Built and integrated a custom debugging system to proactively detect and resolve logic, memory, and hidden runtime errors, ensuring software stability throughout development

Dragonic Tactics | Game Programmer / QA

Present

- Collaborated with a team of 4 to develop a game, contributing as both a programmer and QA member across multiple systems
- Designed and implemented core game systems including Combat, Spell, and AI systems, improving overall gameplay functionality and stability
- Developed a Debug Manager, an in-game tool that streamlined the debugging process for the entire team by centralizing runtime diagnostics and error reporting
- Performed QA testing to identify and triage bugs, working closely with teammates to ensure consistent software quality